

Exception Handling

Study Guide

Mr. Shivkumar Lilhare
Assistant Professor (CSE), PIT
Parul University

1.	Creating User-Defined Exceptions.....	1
2.	Role of JVM in Exception Handling.....	1
3.	Exception Propagation.....	2
4.	Summary of Key Concepts.....	2
5.	References.....	2

8.3.1 Creating User-Defined Exceptions

◆ What is it?

In Java, besides using predefined exceptions, we can create **custom exceptions** by extending the Exception or RuntimeException class.

◆ Syntax:

```
class MyException extends Exception {  
    public MyException(String message) {  
        super(message);  
    }  
}
```

◆ Usage:

```
public class Test {  
    static void validate(int age) throws MyException {  
        if (age < 18)  
            throw new MyException("Not eligible for voting");  
    }  
  
    public static void main(String[] args) {  
        try {  
            validate(15);  
        } catch (MyException e) {  
            System.out.println("Caught: " + e);  
        }  
    }  
}
```

8.3.2 Exception Propagation

◆ What is it?

Exception propagation is the process by which **exceptions move up the call stack** until they are handled or reach the JVM.

◆ Example:

```
void method1() {  
    method2();  
}  
  
void method2() {  
    int result = 10 / 0; // ArithmeticException  
}
```

If not handled in method2, it propagates to method1, and so on.

◆ **Key Point:**

Only **unchecked exceptions** automatically propagate. **Checked exceptions** must be handled explicitly or declared using throws.

8.3.3 Role of JVM in Exception Handling

◆ **What JVM Does:**

- If an exception is **not caught**, the JVM terminates the program.
- Displays the **name of the exception**, **description**, and **stack trace**.
- Example output:

```
Exception in thread "main" java.lang.ArithmeticException: / by zero
    at MyClass.method(MyClass.java:5)
    at MyClass.main(MyClass.java:10)
```

➤ **Summary of Key Concepts:**

- Java allows you to create custom exceptions tailored to your needs.
- Exception propagation means the exception is passed up the method call stack.
- If uncaught, the JVM terminates the program and displays the stack trace.
- Good practices help make code more robust and maintainable.

➤ **References:**

- Oracle Java Documentation □ Creating Custom Exceptions
🔗 <https://docs.oracle.com/javase/tutorial/essential/exceptions/creating.html>
- GeeksforGeeks □ Custom Exceptions in Java
🔗 <https://www.geeksforgeeks.org/user-defined-exceptions-in-java/>
- Baeldung □ Exception Propagation in Java
🔗 <https://www.baeldung.com/java-exception-handling-best-practices>
- JavaTpoint □ Java Exception Propagation
🔗 <https://www.javatpoint.com/exception-propagation>

Parul[®] University | **NAAC** **A++**
Vadodara, Gujarat | **GRADE**



    
<https://paruluniversity.ac.in/>